

Costing, Part B — The build-out: servers, cloud, people, and the levers

Markers as in Part A: [measured] measured · [list price] vendor list price, verify · [derived] derived.

1. Four deployment scales

[derived] Sized from Part A §2 — one GPU ≈ 100,000 artisans, 40% of onboarded artisans active.

	S1 · Pilot	S2 · State	S3 · National	S4 · Full TAM
Artisans onboarded	2,000	250,000	10,00,000	~1 crore
Active (40%)	800	100,000	4,00,000	40,00,000
Photos/month	8,000	10 lakh	40 lakh	4 crore
GPUs needed (full model)	1 (shared)	1 + 1 HA	4 + 2 HA	40 + 8 HA
GPUs with lite	1	1 + 1	3 + 1	26 + 5
App tier (vCPU)	4	32	128	1,024
Postgres	1 managed	HA + 1 replica	HA + 3 replicas + partitioning	Sharded by state
Object storage	30 GB	35 TB	140 TB	1.4 PB

Note the shape. From S1 to S4 artisan count multiplies by **5,000×**; GPU count multiplies by **48×**. **The AI is the sub-linear part of this system.** What scales linearly is storage, bandwidth and — the point of §4 — people.

2. Option A — Buy the servers (capex, government-owned)

The right model for a ministry that wants an asset on its own books, and the cheaper model at sustained high utilisation.

2.1 [warning] The licensing trap, before any price

[list price] **NVIDIA's GeForce driver EULA prohibits datacenter deployment of consumer cards.** An RTX

4090 is roughly a third the price of an L4 and is **not licensable for a government datacenter**. A procurement built on RTX pricing will be rejected at compliance review or, worse, deployed and later found non-compliant.

So all capex below uses datacenter-class cards. Our benchmark numbers came from consumer cards (RTX 4060 / 2050) — [derived] an **L4 is roughly 1.5–2× an RTX 4060** for this workload, so the capacity figures in §1 are **conservative** on L4, which is the direction we want an estimate to err.

2.2 [list price] Per-node hardware, India list — verify with an OEM before quoting

Component	Spec	~₹
GPU × 2	NVIDIA L4 24 GB (72 W, single-slot, no extra cooling)	5,00,000
Server chassis	2U, dual Xeon Silver / EPYC, 24–32 cores	2,50,000
RAM	256 GB ECC	90,000
NVMe	2 × 3.84 TB, RAID 1	90,000
NIC, rails, PSU	25 GbE redundant	40,000
Per GPU node	2 GPUs ⇒ ~2 lakh artisans	≈ ₹9,70,000

[derived] **Amortised over 4 years: ₹20,200/month per node, or ₹10,100 per GPU per month.**

[list price] Running costs per node: power (~450 W under load ⇒ 324 kWh/month at ₹8/unit ≈ **₹2,600**) + colocation 2U with redundant power and 1 Gbps commit (**₹8,000–15,000/month**).

[derived] **All-in per GPU node: ≈ ₹32,000/month.** Serving ~2 lakh artisans ⇒ **₹0.16 per artisan per month** for AI compute.

2.3 [derived] Capex by scale

	S1 Pilot	S2 State	S3 National	S4 Full TAM
GPU nodes	1	1	3	24
App/DB/cache nodes	1	4	12	80
Hardware capex	₹15 L	₹40 L	₹1.6 Cr	₹11 Cr
[derived] Amortised/month (4 yr)	₹31,000	₹83,000	₹3.3 L	₹23 L
Colo + power + bandwidth/month	₹20,000	₹1.2 L	₹4.5 L	₹32 L

	S1 Pilot	S2 State	S3 National	S4 Full TAM
--	----------	----------	-------------	-------------

Total infra/month	₹51,000	₹2.0 L	₹7.8 L	₹55 L
-------------------	---------	--------	--------	-------

Add for a government deployment, and these are not optional: a second site for DR (~60% of primary), MeitY-empanelled cloud or a STQC-audited datacenter, and an annual security audit ([list price] ₹3–8 lakh).

3. Option B — Rent it (opex, cloud)

Right for the pilot, right for burst, and right until utilisation is provably sustained.

[list price] List prices, ap-south-1 / Indian regions — **verify**. Indian MeitY-empanelled alternatives (E2E Networks, Yotta, CtrlS, NxtGen) typically quote **30–50% below hyperscaler list** and satisfy data-localisation procurement conditions — get quotes before assuming AWS pricing.

Line	S1 Pilot	S2 State	S3 National
GPU (T4/L4, 1-yr reserved)	₹20,000	₹40,000	₹2.4 L
App tier (stateless, autoscaled)	₹8,000	₹65,000	₹2.6 L
Postgres (managed, HA + replicas)	₹6,000	₹1.1 L	₹4.0 L
Redis	₹2,000	₹12,000	₹45,000
Object storage (R2)	₹100	₹46,000	₹1.85 L
Egress / CDN (R2 = ₹0 egress)	₹1,700	₹1,700	₹20,000
Monitoring, logs, backups	₹3,000	₹25,000	₹90,000
Total/month	₹41,000	₹3.0 L	₹12.4 L
[derived] Per artisan/month	₹20.50	₹1.20	₹1.24

3.1 [derived] Buy vs rent — the crossover

	Cloud/month	Own/month	Verdict
S1 Pilot	₹41,000	₹51,000	Rent. Utilisation is low and lumpy
S2 State	₹3.0 L	₹2.0 L	Own — if the programme is committed ≥3 years

	Cloud/month	Own/month	Verdict
S3 National	₹12.4 L	₹7.8 L	Own the steady state, rent the peak

[derived] The honest recommendation: hybrid. Own the baseline GPU capacity (utilisation is predictable and high), rent the burst. The enhancement queue is asynchronous by contract, so it tolerates preemption — [list price] spot/preemptible GPU is 60–70% below on-demand and is safe for this workload specifically because `ai/enhance/jobs.py` already treats a lost job as recoverable and rule 3 already guarantees a lost enhancement never costs the listing. An architectural decision taken for reliability turns out to unlock the cheapest compute tier available.

4. The lines that are bigger than the servers

[derived] At S3, infrastructure is ₹7.8–12.4 lakh/month. **These three are larger.**

4.1 [blocking] GeM order reconciliation — the largest cost in the entire programme

GeM has no order API. Orders appear on the GeM seller dashboard and nowhere else, so a human reads it and marks the order in the admin console (`/gem-recon`).

[derived] Assume 25% of active artisans transact on GeM, and one coordinator handles 500 such artisans:

	S2 State	S3 National
GeM-active artisans	25,000	100,000
Coordinators needed	50	200
[list price] At ₹25,000/month each	₹12.5 L/month	₹50 L/month
Per year	₹1.5 Cr	₹6 Cr

[derived] A GeM order-notification API — one MoU, zero engineering on their side beyond what already exists — removes ₹6 crore a year of recurring public expenditure.

This is the single most valuable thing to put in front of a ministry panel. It costs them nothing, it is entirely within their gift, and it converts a permanent operating expense into a webhook. We should ask for it in the room.

And the counter-argument, stated fairly: those 200 coordinator posts are 200 rural jobs, and the same person also runs the Cluster Hub — quality check, packing, consolidated pickup. **The role does not disappear if the API arrives; the reconciliation half of it does,** which is the half that produces no value for anyone.

4.2 [open] ECO compliance — TCS and monthly GSTR-8

Because we become the ONDC Marketplace Seller Node, **we** are the e-commerce operator. Per **CBIC Circular 194/06/2023**, where the supplier-side ECO is not itself the supplier, **it carries the TCS liability**. That means TCS collection at 0.5% CGST + 0.5% SGST on net taxable supplies, and **monthly GSTR-8 filing**.

[list price] Compliance retainer: ₹40,000–₹1,00,000/month at S2, scaling to **₹3–5 lakh/month at S3** plus reconciliation tooling. [derived] **₹35–60 lakh a year at national scale**.

This is the price of Tier A and it is worth it — it is the mechanism by which an artisan with no GST registration can sell nationally. But it is a real, recurring, non-technical obligation and a costing that omits it is wrong.

4.3 [open] DPDP and security

Line	[list price] Annual
Named Data Protection Officer (1 FTE at S3)	₹18–25 L
Consent-manager integration + audit	₹8–15 L
Annual security audit / VAPT (STQC or CERT-In empanelled)	₹3–8 L
Breach-response retainer	₹5–10 L
Total at S3	₹35–58 L/year

Non-negotiable, not discretionary. DPDP penalties reach **₹250 crore** for serious violations and **₹200 crore** for failure to notify a breach. [derived] **The entire compliance budget is 0.2% of a single maximum penalty.**

4.4 [derived] Engineering and support

	S2 State	S3 National
Engineers (backend, app, ML, infra)	5	12
[list price] At ₹18 L/year loaded	₹90 L	₹2.16 Cr
Support / helpline (multilingual)	4	20
[list price] At ₹4.5 L/year	₹18 L	₹90 L
Total/year	₹1.08 Cr	₹3.06 Cr

5. [derived] Total cost of ownership — the number a ministry actually needs

	S1 Pilot	S2 State	S3 National
Artisans onboarded	2,000	2,50,000	10,00,000
Infrastructure	₹6 L	₹24 L	₹94 L
Marginal (LLM/ASR/storage), with Bhashini	₹0.6 L	₹65 L	₹2.6 Cr
GeM reconciliation	₹6 L	₹1.5 Cr	₹6.0 Cr
ECO compliance	₹3 L	₹12 L	₹50 L
DPDP + security	₹5 L	₹25 L	₹50 L
Engineering + support	₹60 L	₹1.08 Cr	₹3.06 Cr
TOTAL / YEAR	≈ ₹81 L	≈ ₹3.84 Cr	≈ ₹13.6 Cr
[derived] Per artisan / year	₹4,050	₹154	₹136
[derived] Per artisan / year, if GeM ships an order API	₹3,750	₹94	₹76

Read the last two rows together. Cost per artisan falls by 96% from pilot to state scale and then flattens — because the fixed engineering and compliance base is spread wider while the variable lines stay tiny. This system gets dramatically cheaper per head as it grows, which is exactly the property a national programme needs and exactly what per-image API pricing would have destroyed.

5.1 The comparison that frames it

[list price] Context figures — verify against current budget documents before quoting:

Reference	Scale
PM Vishwakarma scheme outlay	~₹13,000 crore
[derived] This system, national, 10 lakh artisans, per year	₹13.6 crore
[derived] As a share of that outlay	≈ 0.1%

[derived] One-tenth of one percent of an existing scheme's outlay digitises ten lakh artisans — permanently, with a catalog they own, on channels the government already runs.

6. Cost-reduction levers, ranked by rupees saved

The ministry will ask how to spend less. This is the answer, in order.

#	Lever	[derived] Annual saving at S3	Effort
1	Settle speech pricing — Sarvam volume rate, or Bhashini free tier	₹18 Cr avoided	Two procurement calls. Swap surface is 2 functions
2	Self-host CV (already decided) — vs a per-image API	₹17.6 Cr avoided	Done
3	GeM order API MoU	₹6.0 Cr	A ministry conversation, no code
4	Self-host the description LLM past crossover (§6.1)	₹1.4 Cr	~2 weeks
5	Cloudflare R2 instead of S3 — zero egress	₹1.2 Cr	One afternoon
6	Pipeline CPU/GPU separately — 3.2× throughput	~₹25 L, and defers GPU purchases	A refactor
7	BiRefNet-lite — 1.47× on top, half the VRAM	~₹15 L	A config flag. Already benchmarked
8	Spot/preemptible GPU for the enhance queue	60–70% of burst compute	Safe by existing design
9	Archive originals after 90 days	~₹10 L	A lifecycle rule
10	Client-side downscale before upload	Artisan's own data cost, and our ingress	The single highest-value unbuilt app item

Levers 1, 3 and 5 together are worth ₹25 crore a year and none of them is an engineering problem. Two are conversations and one is a vendor choice. That is the costing insight worth saying out loud in the room.

6.1 [derived] The self-hosted-LLM crossover, worked

Description generation costs **₹1.66/product** on a frontier API (Part A §3.2). A self-hosted open-weights model (an Indic-capable 7–12B, served on a GPU we already own) costs [derived] roughly **₹0.05/product** in amortised compute, plus one dedicated GPU node at ₹32,000/month.

```
Crossover: monthly_products × (₹1.66 - ₹0.05) > ₹32,000
monthly_products > 19,900
```

[derived] **Past ~20,000 products a month — reached at roughly 6,700 active artisans — self-hosting the description model is cheaper.** S2 and S3 are both far past it.

The caveat that decides whether to actually do it: quality. EN + HI SEO descriptions plus structured field extraction is the hardest language task in the system, and a cheaper model that produces a rejected listing has saved nothing. **The right sequencing is: keep the API until we have a held-out evaluation set from real listings, then migrate on measured quality, not on the crossover date.** [key] And an Indian open-weights model here is both the cheaper answer and the better political one on a government deployment — which is the honest argument for putting Sarvam on the slide *after* wiring it, not before (Part 0, A0.4).

7. Cost questions a judge will ask

Q — "Your AI costs will explode when you scale." [derived] No — they are sub-linear and we can show why. The model that runs on **every** photograph is self-hosted, so its marginal cost is electricity: **₹0.005 per photograph**. One GPU serves ~100,000 artisans. The per-image API alternative would cost **₹1.76 per photograph — 345× more** — and that is the decision that determines whether this scales, not the model choice.

Q — "What is the cheapest way to run this?" Own the baseline GPUs, rent the burst on preemptible instances, Cloudflare R2 for storage and egress, Bhashini for speech, self-host the description model past ~20,000 products/month. [derived] That combination takes national cost from ~₹13.6 crore to **~₹6 crore a year** — before the GeM API, which takes it to **~₹5.5 crore**.

Q — "What if the government wants everything on Indian soil?" It already almost is, and it makes us *cheaper*, not dearer. Self-hosted CV runs anywhere we put it. Postgres, Redis and object storage go to a MeitY-empanelled Indian provider — [list price] typically **30–50% below hyperscaler list**. The **only** cross-border dependency is the LLM API, and §6.1 shows we should self-host it past 20,000 products/month anyway. **Data localisation and cost reduction point the same direction here**, which is unusual and worth saying.

Q — "What happens if funding stops?" [derived] The artisan's catalog is portable by design — products live in our DB and export to GeM Excel, ONDC, and platform payloads. **The minimum survivable configuration is read-only serving of existing catalogs plus GeM export: one small GPU node and one app node, ~₹60,000/month.** New enhancement stops; nothing already published disappears. **That is a deliberate property of the factory model** (docs/Master-Technical-Reference.md §2): *the catalog is ours, and edit once → all exports regenerate*.

Q — "Your per-artisan cost is ₹136. Prove it is not ₹1,360." [derived] The number rests on three things, and only one is uncertain. **GPU capacity is measured** (402 ms, 1615 MiB, on named hardware). **Storage and egress are vendor list prices.** The uncertain one is **ASR** — Bhashini free versus commercial is the difference between ₹136 and ₹316 per artisan-year, and it is docs/decisions.md #5, still open. **We have given you both numbers rather than the flattering one.**

8. [do not] What must be verified before any of this is quoted in a procurement conversation

#	Item	Owner
1	[blocking] Bhashini commercial terms, rate limits, eligibility — the single largest cost variable	
2	[blocking] NVIDIA L4 / L40S India OEM quotes, and datacenter licensing confirmation	
3	[blocking] MeitY-empanelled provider quotes (E2E, Yotta, CtrlS, NxtGen) vs hyperscaler list	
4	[open] Current OpenRouter / Anthropic list prices — they move	
5	[open] Cloudflare R2 India egress and any regional caveats	
6	[open] GeM Vendor Assessment (~₹11,200) — are artisan OEMs exempt? In writing	
7	[open] ONDC network participant fees and any transaction levy	
8	[open] Coordinator salary norms under existing cluster schemes — §4.1 rests on ₹25,000/month	
9	[open] PM Vishwakarma outlay, from the current budget document, for the §5.1 comparison	

*Same rule as research/pricing: a **dated vendor quote is research**. A **remembered price is a guess**. Every [list price] in this document is a remembered price until someone replaces it with a quote.*